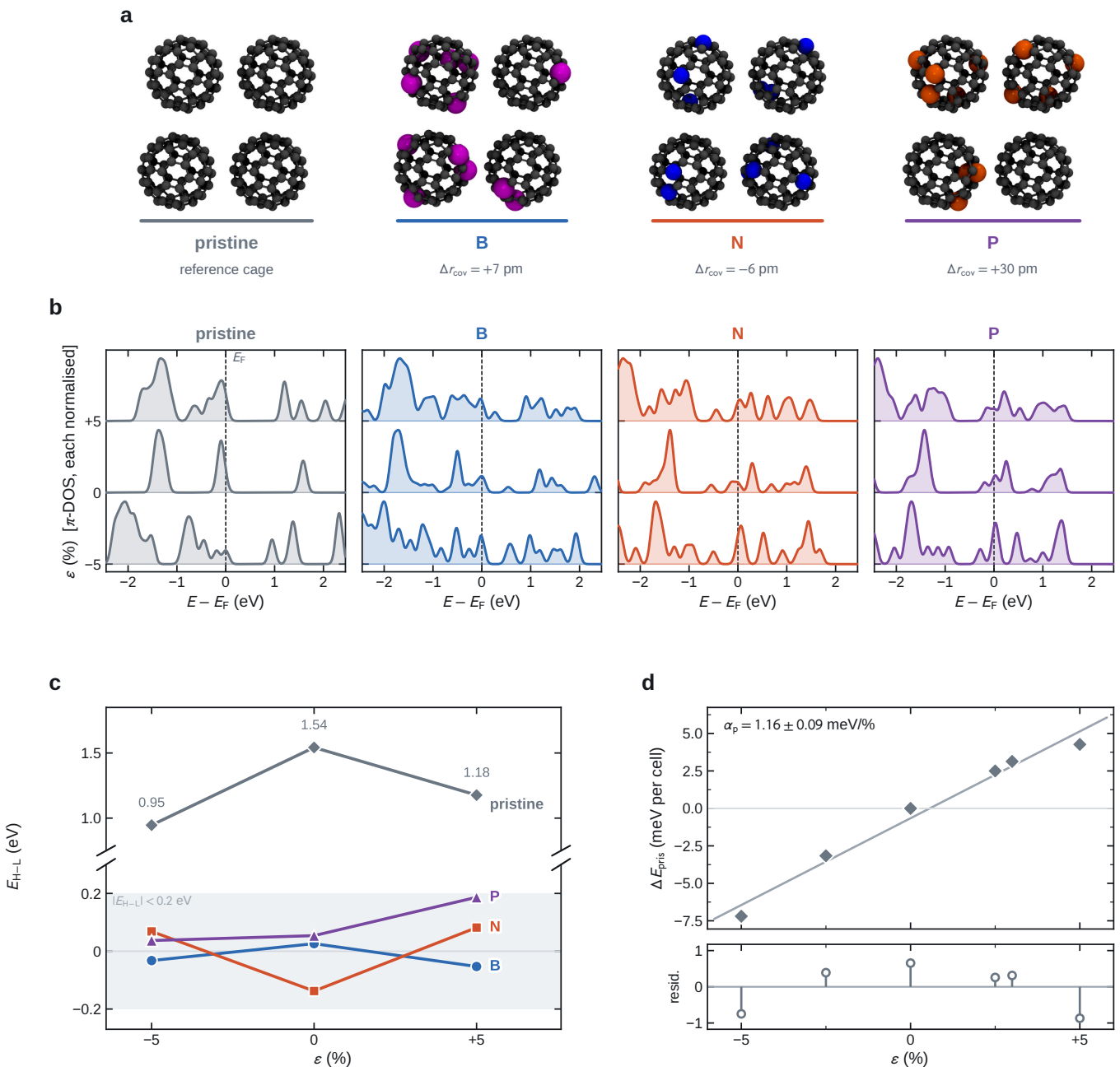

a
pristine
reference cage
B
Δr_cov = +7 pm
N
Δr_cov = −6 pm
P
Δr_cov = +30 pm
b
pristine
B
N
P
E_F
ε (%) [π-DOS, each normalised]
+5
0
−5
−2
−1
0
1
2
E − E_F (eV)
c
1.54
1.18
0.95
pristine
|E_H−L| < 0.2 eV
P
N
B
E_H−L (eV)
1.5
1.0
0.2
0
−0.2
−5
0
+5
ε (%)
d
α_p = 1.16 ± 0.09 meV/%
ΔE_pris (meV per cell)
5.0
2.5
0.0
−2.5
−5.0
−7.5
resid.
1
0
−1
−5
0
+5
ε (%)